NEET UG HIGH-YIELD CHEMISTRY

Solutions: Principles, Equilibria & Colligative Properties

A Comprehensive Exam-Oriented Master Tutorial and Rigorous Question Bank

Core Objectives for NEET UG Aspirants

By mastering this tutorial, you will be able to:

- Identify and classify all nine binary solution types according to the physical states of solute and solvent.
- Interconvert concentration expressions (mass %, volume %, ppm, mole fraction, molarity, molality) and predict temperature dependence.
- Formulate and apply **Henry's Law** ($p = K_{\mathrm{H}} x$), understanding the physiological relevance in aquatic life, scuba diving (bends), and high altitude (anoxia).
- Apply **Raoult's Law** for liquid-liquid volatile mixtures and calculate vapour phase compositions (y_i).
- Differentiate between **ideal solutions** ($\Delta_{\mathrm{mix}} H = 0$, $\Delta_{\mathrm{mix}} V = 0$) and non-ideal solutions exhibiting positive and negative deviations, including azeotropic compositions.
- Derive and solve numerical problems on the four **colligative properties**: Relative Lowering of Vapour Pressure (RLVP), Boiling Point Elevation (ΔT_{b}), Freezing Point Depression (ΔT_{f}), and Osmotic Pressure (Π).
- Evaluate abnormal molar masses through the **van 't Hoff factor** (i) for dissociating and associating solutes.

1. Types of Solutions and Physical States

A **solution** is defined as a homogeneous mixture of two or more chemically non-reacting substances whose composition and properties are completely uniform throughout. In a **binary solution**, two components are present:

- **Solvent:** The component present in the largest physical quantity, which determines the overall physical state of the solution.
- **Solute:** The minor component(s) dispersed uniformly at molecular or ionic dimensions within the solvent.

Tutorial Table 1: Classification of Binary Solutions

Solution Type	Solute State	Solvent State	High-Yield Textbook & Exam Examples
Gaseous Solutions	Gas	Gas	Mixture of oxygen ($\mathrm{O_2}$) and nitrogen ($\mathrm{N_2}$) gases (air)
	Liquid	Gas	Chloroform ($\mathrm{CHCl_3}$) vapours mixed with nitrogen gas; humidity
	Solid	Gas	Camphor vapours in nitrogen gas; naphthalene in air
Liquid Solutions	Gas	Liquid	Oxygen or carbon dioxide dissolved in water
	Liquid	Liquid	Ethanol ($\mathrm{C_2H_5OH}$) dissolved in water
	Solid	Liquid	Glucose ($\mathrm{C_6H_{12}O_6}$) or sodium chloride (NaCl) in water
Solid Solutions	Gas	Solid	Solution of hydrogen gas adsorbed in palladium ($\mathrm{H_2/Pd}$)
	Liquid	Solid	Amalgam of mercury with sodium ($\mathrm{Na - Hg}$)
	Solid	Solid	Copper dissolved in gold; brass ($\mathrm{Cu + Zn}$), bronze ($\mathrm{Cu + Sn}$)

NEET Memory Insight: Note that brass is an alloy of copper and zinc, German silver contains copper, zinc, and nickel, while bronze is an alloy of copper and tin. Notice that German silver contains *no silver*! In biological fluids, a fluoride ion concentration of $1.0\ \mathrm{ppm}$ prevents dental caries, $1.5\ \mathrm{ppm}$ causes permanent enamel mottling, and higher concentrations are lethal poisons (e.g., NaF in rodenticides).

2. Quantitative Expressions of Solution Concentration

The concentration of a solution governs its chemical reactivity and colligative behaviour. The standard quantitative units and their mathematical definitions are summarized below.

BOSONANCE

Summary of Concentration Formulations

- **Mass Percentage (\% w/w):**

$$\text{Mass } \% \text{ of component} = \frac{\text{Mass of component in solution}}{\text{Total mass of solution}} \times 100$$

Example: Commercial bleach contains 3.62% (w/w) sodium hypochlorite (NaOCl) in water.

- **Volume Percentage (\% v/v):**

$$\text{Volume } \% \text{ of component} = \frac{\text{Volume of component}}{\text{Total volume of solution}} \times 100$$

Example: A 35% (v/v) aqueous solution of ethylene glycol is used as an automotive engine coolant/antifreeze; it lowers the freezing point of water to $255.4\,\mathrm{K}$ ($-17.6^{\circ}\mathrm{C}$).

- **Mass by Volume Percentage (\% w/v):**

$$\text{Mass by volume } \% = \frac{\text{Mass of solute in grams}}{\text{Total volume of solution in mL}} \times 100$$

Extensively used in pharmacology and medicine (e.g., normal saline is $0.9\%\ \mathrm{w/v\ NaCl}$).

- **Parts Per Million (ppm):** Used for trace constituents and environmental pollutants:

$$\text{ppm} = \frac{\text{Number of parts of the component}}{\text{Total number of parts of all components}} \times 10^{6}$$

- **Mole Fraction (x):** Ratio of moles of a given component to the total moles of all species present. For a binary mixture of components A and B:

$$x_A = \frac{n_A}{n_A + n_B}, \quad x_B = \frac{n_B}{n_A + n_B}, \quad x_A + x_B = 1$$

- **Molarity (M):** Number of moles of solute dissolved per litre (dm^3) of solution:

$$M = \frac{\text{Moles of solute } (n_2)}{\text{Volume of solution in litres } (V)} = \frac{w_2 \times 1000}{M_2 \times V\,(\mathrm{mL})} \quad [\mathrm{mol\,L^{-1}}]$$

- **Molality (m):** Number of moles of solute dissolved per kilogram of pure solvent:

$$m = \frac{\text{Moles of solute } (n_2)}{\text{Mass of solvent in kg } (w_1/1000)} = \frac{w_2 \times 1000}{M_2 \times w_1\,(\mathrm{g})} \quad [\mathrm{mol\,kg^{-1}}]$$

Crucial NEET Distinction — Temperature Dependence: Mass does not alter with thermal expansion or contraction. Consequently, **mass percentage, ppm, mole fraction, and molality (m) are temperature-independent**. In contrast, **molarity (M) and mass-by-volume percentage (\% w/v) depend directly on temperature** because liquid volume expands upon heating, decreasing the observed molarity.

BOSONANCE

WORKED EXAMPLE: MOLARITY, MOLALITY, AND MOLE FRACTION INTERCONVERSION

Problem: Calculate the molality, molarity, and mole fraction of KI in an aqueous solution containing 20% KI by mass, given the solution density is $1.202\ \mathrm{g\,mL^{-1}}$. (Molar mass of $\mathrm{K} = 39\ \mathrm{g\,mol^{-1}}$, $\mathrm{I} = 127\ \mathrm{g\,mol^{-1}}$; $\mathrm{KI} = 166\ \mathrm{g\,mol^{-1}}$).

Step-by-Step Solution:

1. **Basis:** Consider $100\ \mathrm{g}$ of solution.
   Mass of KI (w_2) = $20\ \mathrm{g}$.
   Mass of solvent water (w_1) = $100 - 20 = 80\ \mathrm{g} = 0.080\ \mathrm{kg}$.
2. **Moles of solute:**
$$n_{\mathrm{KI}} = \frac{20\ \mathrm{g}}{166\ \mathrm{g\,mol^{-1}}} = 0.1205\ \mathrm{mol}$$
3. **Molality (m):**
$$m = \frac{n_{\mathrm{KI}}}{\text{Mass of water (kg)}} = \frac{0.1205\ \mathrm{mol}}{0.080\ \mathrm{kg}} = 1.506\ \mathrm{mol\,kg^{-1}} \approx 1.51\ m$$
4. **Molarity (M):**
$$\text{Volume of } 100\ \mathrm{g} \text{ solution} = \frac{\text{Mass}}{\text{Density}} = \frac{100\ \mathrm{g}}{1.202\ \mathrm{g\,mL^{-1}}} = 83.19\ \mathrm{mL} = 0.08319\ \mathrm{L}$$
$$M = \frac{0.1205\ \mathrm{mol}}{0.08319\ \mathrm{L}} = 1.448\ \mathrm{mol\,L^{-1}} \approx 1.45\ M$$
5. **Mole Fraction (x_{KI}):**
$$n_{\mathrm{H_2O}} = \frac{80\ \mathrm{g}}{18.0\ \mathrm{g\,mol^{-1}}} = 4.444\ \mathrm{mol}$$
$$x_{\mathrm{KI}} = \frac{n_{\mathrm{KI}}}{n_{\mathrm{KI}} + n_{\mathrm{H_2O}}} = \frac{0.1205}{0.1205 + 4.444} = \frac{0.1205}{4.5645} = 0.0264$$

3. Solubility Equilibria & Gas-Liquid Systems

Solubility is the maximum amount of solute that dissolves in a specified quantity of solvent at a designated temperature and pressure to establish dynamic equilibrium with the undissolved solute:

$$\mathrm{Solute} + \mathrm{Solvent} \rightleftharpoons \mathrm{Solution}$$

When the forward rate of dissolution equals the backward rate of crystallisation, the solution is **saturated**. If additional solute can dissolve, it is **unsaturated**.

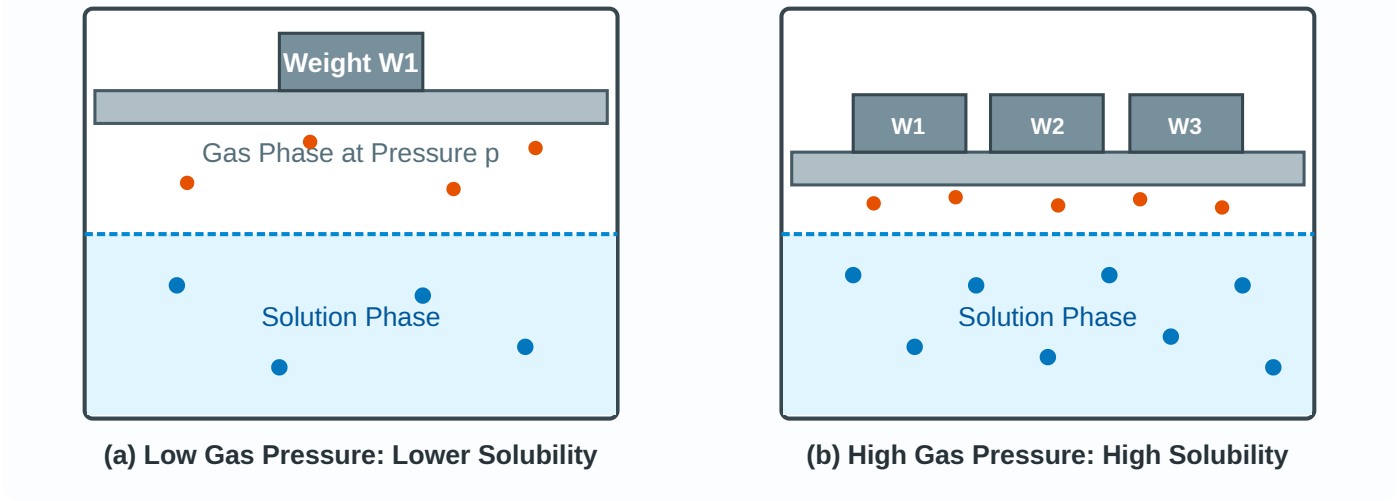
3.1 Solubility of Solids in Liquids

- **Nature of Solute and Solvent:** Polar solutes (NaCl, sugar) dissolve in polar solvents ($\mathrm{H_2O}$); non-polar solutes (naphthalene, anthracene) dissolve readily in non-polar solvents (benzene, $\mathrm{CS_2}$). *"Like dissolves like."*
- **Effect of Temperature:** According to Le Chatelier's principle, if dissolution is endothermic ($\Delta_{\mathrm{sol}}H > 0$), solubility rises with increasing temperature. If dissolution is exothermic ($\Delta_{\mathrm{sol}}H < 0$), solubility decreases as temperature rises.
- **Effect of Pressure:** Negligible, because condensed phases (liquids and solids) are virtually incompressible.

3.2 Solubility of Gases in Liquids & Henry's Law

Unlike solids, gases are compressible. Compressing a gas increases its concentration above the liquid, accelerating molecular bombardment upon the liquid interface until a higher dissolved equilibrium concentration is reached.

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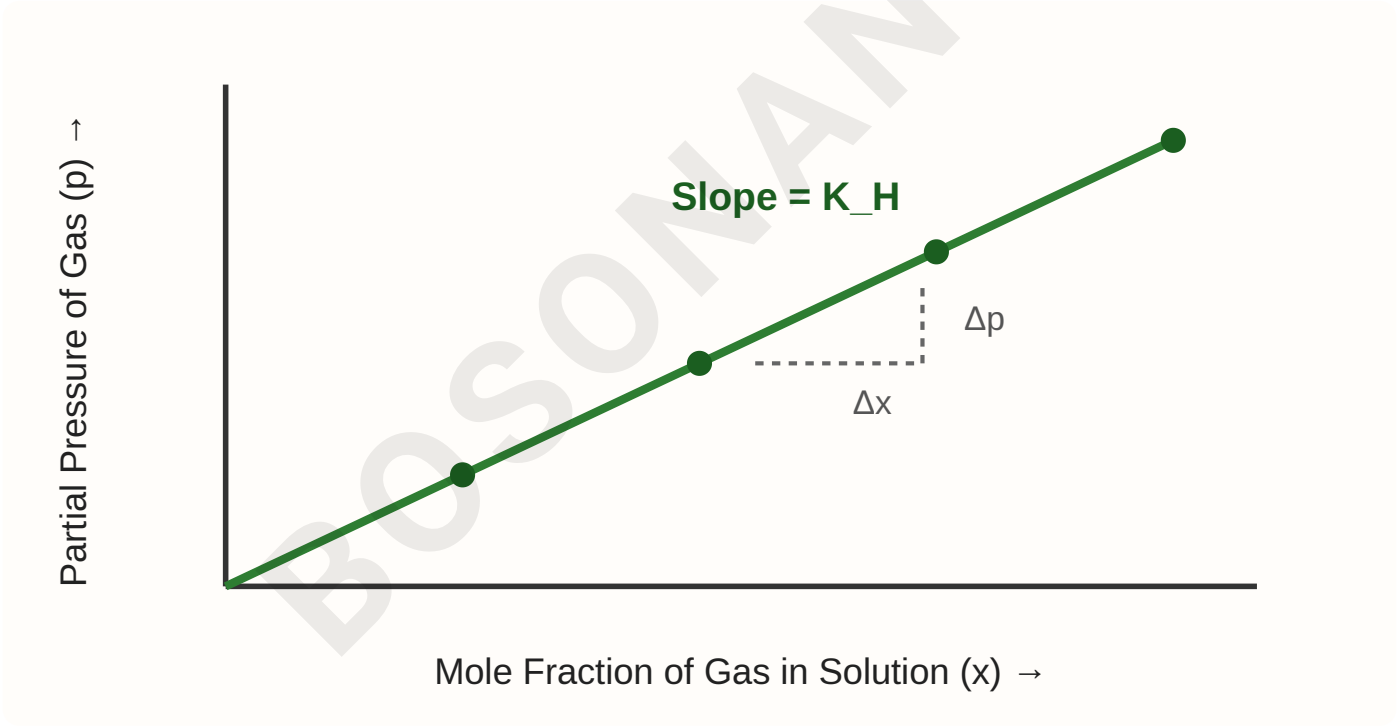


Tutorial Diagram 1: Molecular model showing how compression of gas particles increases the rate of collision and dynamic solubility.

Henry's Law Statement: At constant temperature, the solubility of a gas in a liquid is directly proportional to the partial pressure of the gas above the liquid. Formulating solubility in terms of mole fraction (x):

$$p = K_{\mathrm{H}}\, x$$

Where p is the partial pressure of the gas, x is its mole fraction in solution, and K_{H} is the **Henry's Law Constant**.



Tutorial Diagram 2: Linear relationship between partial pressure (p) and mole fraction (x) for a gas obeying Henry's Law.

Key Characteristics of K_{H} for NEET UG:

- **Inverse Solubility Dependence:** At a fixed partial pressure p, since $x = p/K_{\mathrm{H}}$, **a higher K_{H} value signifies lower solubility**.
- **Effect of Temperature on K_{H}:** For gases such as $\mathrm{N_2}$ and $\mathrm{O_2}$, K_{H} increases with rising temperature. Therefore, gas solubility decreases as temperature increases.
- **Biological Ramification:** Aquatic species (fishes, aerobic microbes) are far more comfortable in cold water than warm water because dissolved oxygen concentrations are significantly higher at lower temperatures.

Tutorial Table 2: Henry's Law Constants (K_{H}) for Common Gases in Water

Gas	Temperature (K)	K_{H} (kbar)	Gas	Temperature (K)	K_{H} (kbar)
He	293	144.97	Ar	298	40.3
$\mathrm{H_2}$	293	69.16	$\mathrm{CO_2}$	298	1.67
$\mathrm{N_2}$	293	76.48	$\mathrm{CH_4}$	298	0.413

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Gas	Temperature (K)	K_H (kbar)	Gas	Temperature (K)	K_H (kbar)
N_2	303	88.84	HCHO	298	1.83×10^{-5}
O_2	293	34.86	Vinyl Chloride	298	0.611
O_2	303	46.82	–	–	–

Biological and Industrial Applications of Henry's Law:

1. **Carbonated Beverages:** Soft drink and soda bottles are sealed under high pressure to force larger quantities of CO_2 to dissolve.
2. **Scuba Diving and Decompression Sickness ("Bends"):** As a scuba diver descends, hydrostatic pressure increases, driving atmospheric N_2 into arterial blood. Upon rapid ascent, ambient pressure drops abruptly, decreasing N_2 solubility and causing N_2 gas bubbles to nucleate in blood vessels. These bubbles block capillaries, creating a dangerous and painful condition termed **the bends**. To avert this, scuba tanks are charged with compressed air diluted with helium (11.7% He, 56.2% N_2, 32.1% O_2). Helium has exceptionally low solubility in blood (very high K_H).
3. **High-Altitude Hypoxia ("Anoxia"):** At elevated mountain altitudes, atmospheric pressure is low, lowering O_2 partial pressure. This causes low blood oxygen levels in mountaineers, inducing fatigue, disorientation, and impaired cognition (anoxia).
4. **Thermodynamic Basis:** Dissolution of a gas resembles a condensation phase change: $\text{Gas} + \text{Liquid} \rightleftharpoons \text{Solution} + \text{Heat}$ ($\Delta_{sol}H < 0$). Since the dissolution is exothermic, Le Chatelier's principle dictates that increasing temperature shifts the equilibrium backward, releasing dissolved gas.

4. Vapour Pressure and Raoult's Law

At dynamic equilibrium between evaporation and condensation in a closed container, the pressure exerted by the vapour phase over the liquid phase is called the **vapour pressure** of the liquid.

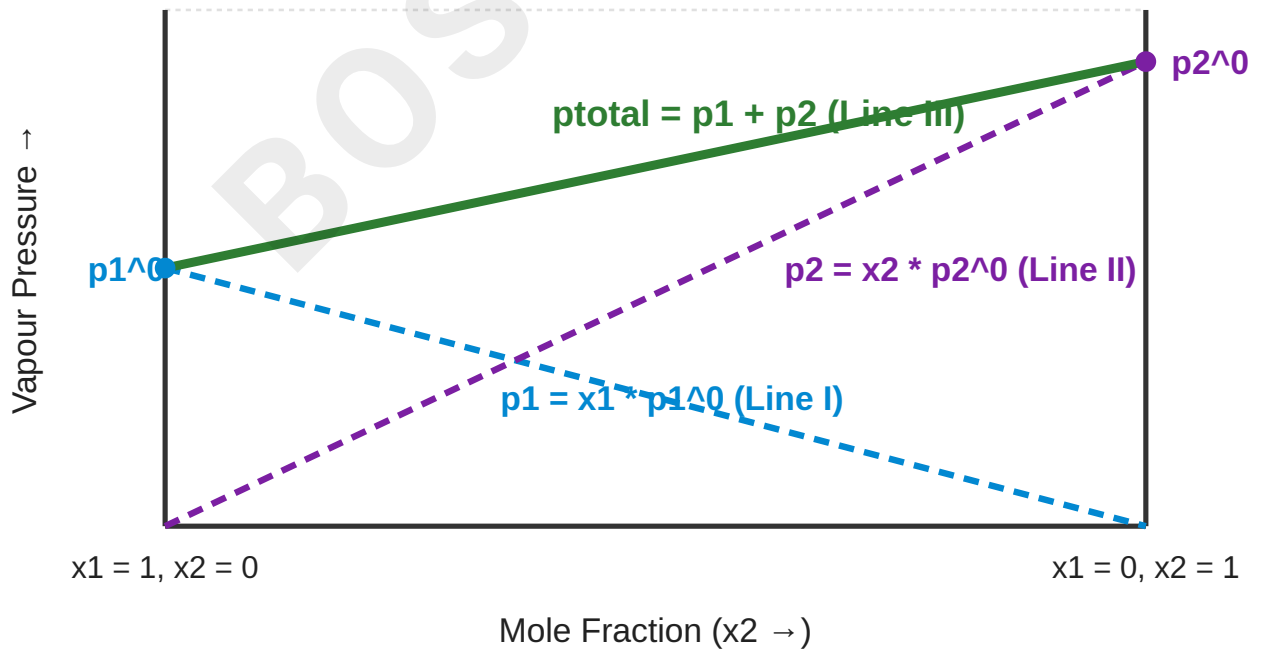
4.1 Binary Solutions of Two Volatile Liquids

Raoult's Law: For a solution of volatile liquids, the partial vapour pressure of each volatile component is directly proportional to its mole fraction in the solution:

$$p_1 = x_1\, p_1^\circ \quad \text{and} \quad p_2 = x_2\, p_2^\circ$$

Applying Dalton's Law of Partial Pressures, the total pressure p_{total} is:

$$p_{\text{total}} = p_1 + p_2 = x_1 p_1^\circ + x_2 p_2^\circ = (1 - x_2) p_1^\circ + x_2 p_2^\circ = p_1^\circ + (p_2^\circ - p_1^\circ) x_2$$



Tutorial Diagram 3: Vapour pressure vs. composition plot for an ideal binary solution obeying Raoult's Law.

4.2 Composition of the Vapour Phase

If y_1 and y_2 represent the mole fractions of components 1 and 2 in the **vapour phase**, Dalton's law dictates:

BOSONANCE

$$p_1 = y_1\, p_{\text{total}} \implies y_1 = \frac{p_1}{p_{\text{total}}} \quad \text{and} \quad y_2 = \frac{p_2}{p_{\text{total}}}$$

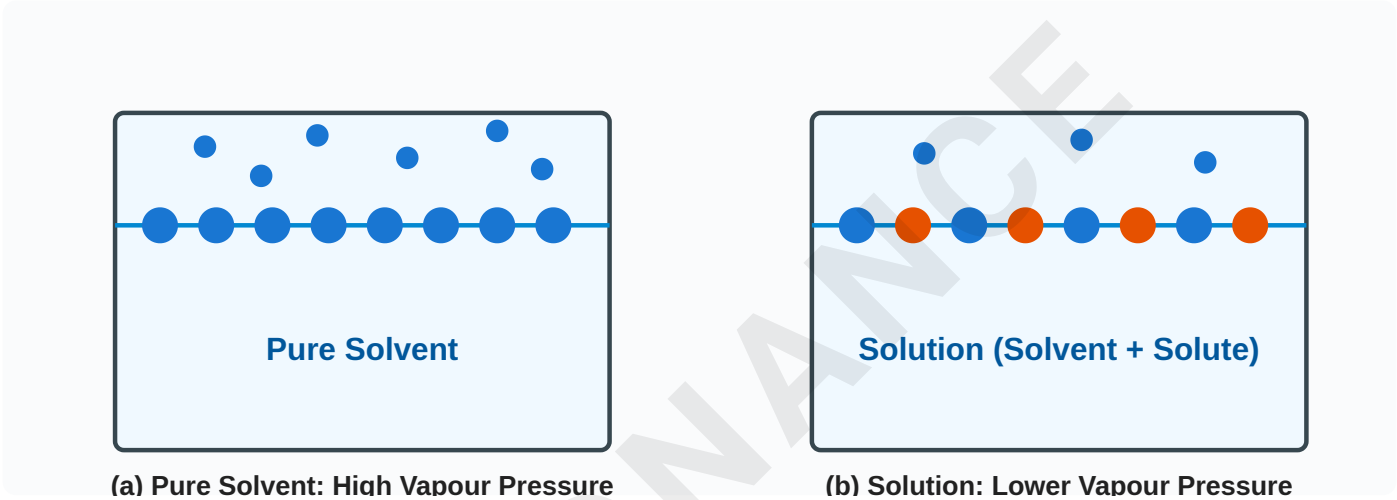
Konovalov's Rule / High-Yield Rule: At equilibrium, the vapour phase is always richer in the component that is more volatile (i.e., having the higher pure vapour pressure p°).

4.3 Raoult's Law as a Special Case of Henry's Law

Comparing both equations:
• Raoult's Law: $p = x\, p^\circ$
• Henry's Law: $p = x\, K_{\text{H}}$
Both laws state that the partial vapour pressure of a volatile species is proportional to its mole fraction in solution. They differ only in the proportionality constant: when $K_{\text{H}} = p^\circ$, Henry's law becomes identical to Raoult's law.

4.4 Solutions of Non-Volatile Solids in Liquids

When a non-volatile solute is dissolved in a solvent, solute particles occupy a fraction of the surface liquid layer. Because non-volatile solute particles do not evaporate, the surface density of escaping solvent molecules is diminished, causing a proportional **lowering of vapour pressure**.



Tutorial Diagram 4: Surface blockage mechanism. Non-volatile solute particles (orange) reduce the solvent escaping surface area.

5. Ideal and Non-Ideal Solutions & Azeotropes

5.1 Ideal Solutions

An **ideal solution** obeys Raoult's Law over the entire range of concentration and temperature.

- **Thermodynamic Criteria:**
$$\Delta_{\text{mix}} H = 0 \quad \text{and} \quad \Delta_{\text{mix}} V = 0$$
(Additionally, for all spontaneous mixing, $\Delta_{\text{mix}} S > 0$ and $\Delta_{\text{mix}} G < 0$).
- **Intermolecular Forces:** The attractive forces between solute-solvent molecules (A--B) are identical in strength to the pure solvent-solvent (A--A) and solute-solute (B--B) interactions.
- **Classic Examples:**
  • n-hexane and n-heptane
  • Benzene and toluene
  • Bromoethane and chloroethane

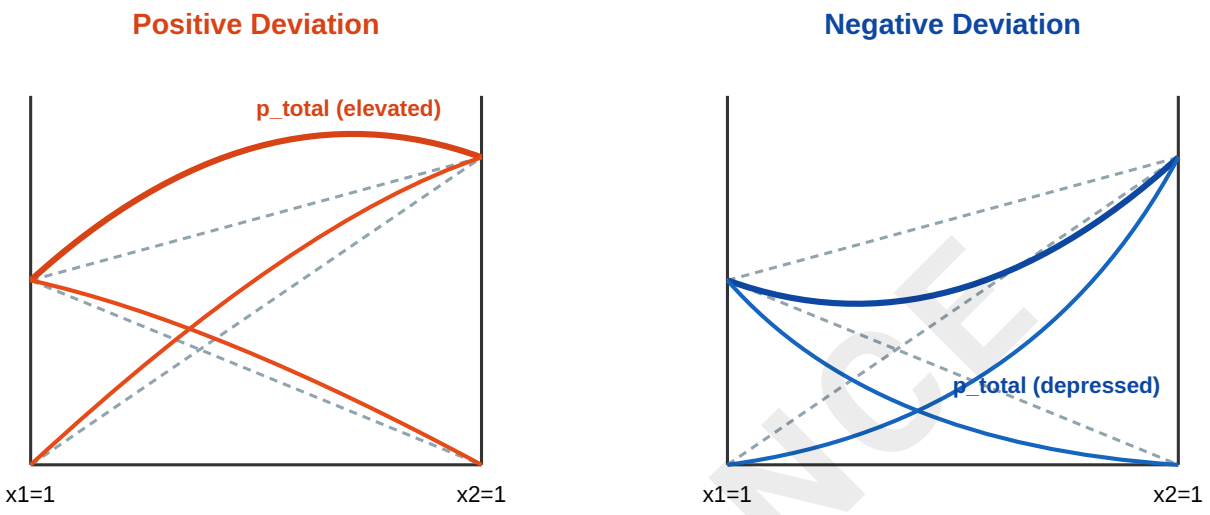
5.2 Non-Ideal Solutions: Positive and Negative Deviations

Tutorial Table 3: Comparison of Positive vs. Negative Deviations from Raoult's Law

Property / Criterion	Positive Deviation ($p_{\text{obs}} > p_{\text{calc}}$)	Negative Deviation ($p_{\text{obs}} < p_{\text{calc}}$)
Molecular Interactions	A--B interactions are **weaker** than A--A and B--B	A--B interactions are **stronger** than A--A and B--B
Escaping Tendency	Higher escaping tendency for both components into vapour phase	Lower escaping tendency; molecules are trapped by strong new bonds
Observed Vapour Pressure	$p_1 > x_1 p_1^\circ, p_2 > x_2 p_2^\circ, p_{\text{total}} > x_1 p_1^\circ + x_2 p_2^\circ$	$p_1 < x_1 p_1^\circ, p_2 < x_2 p_2^\circ, p_{\text{total}} < x_1 p_1^\circ + x_2 p_2^\circ$
$\Delta_{\text{mix}} H$	$\Delta_{\text{mix}} H > 0$ (Endothermic mixing; heat is absorbed)	$\Delta_{\text{mix}} H < 0$ (Exothermic mixing; heat is evolved)
$\Delta_{\text{mix}} V$	$\Delta_{\text{mix}} V > 0$ (Expansion upon mixing)	$\Delta_{\text{mix}} V < 0$ (Contraction upon mixing)

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Property / Criterion	Positive Deviation ($p_{\text{obs}} > p_{\text{calc}}$)	Negative Deviation ($p_{\text{obs}} < p_{\text{calc}}$)
Azeotrope Formed	**Minimum Boiling Azeotrope** (at a specific composition, boiling point is lower than both pure components)	**Maximum Boiling Azeotrope** (at a specific composition, boiling point is higher than both pure components)
Textbook Examples	• Ethanol + Acetone (acetone disrupts ethanol H-bonds) • Acetone + CS_2 • Water + Ethanol (95% ethanol azeotrope)	• Chloroform + Acetone (forms strong $\text{C--H}\cdots\text{O}$ H-bond) • Phenol + Aniline (phenolic $-\text{OH}$ proton H-bonds to aniline $-\text{NH}_2$) • Nitric acid (68%) + Water (32%) ($\text{b.p.} = 393.5\,\text{K}$)



Tutorial Diagram 5: Vapour pressure plots showing positive deviation (elevated curves) and negative deviation (depressed curves) relative to Raoult's Law (dashed).

6. Colligative Properties & Molar Mass Determination

Colligative properties (Latin: *co* = together, *ligare* = to bind) depend **strictly on the total number of solute particles relative to total molecules present** in solution, and **never on the chemical nature, shape, or identity of the solute**.

6.1 Relative Lowering of Vapour Pressure (RLVP)

For a solution of a non-volatile solute in a volatile solvent, by Raoult's law:

$$p_1 = x_1 p_1^\circ = (1 - x_2) p_1^\circ \implies p_1^\circ - p_1 = x_2 p_1^\circ$$

Dividing by p_1° yields the **Relative Lowering of Vapour Pressure**:

$$\frac{\Delta p_1}{p_1^\circ} = \frac{p_1^\circ - p_1}{p_1^\circ} = x_2 = \frac{n_2}{n_1 + n_2}$$

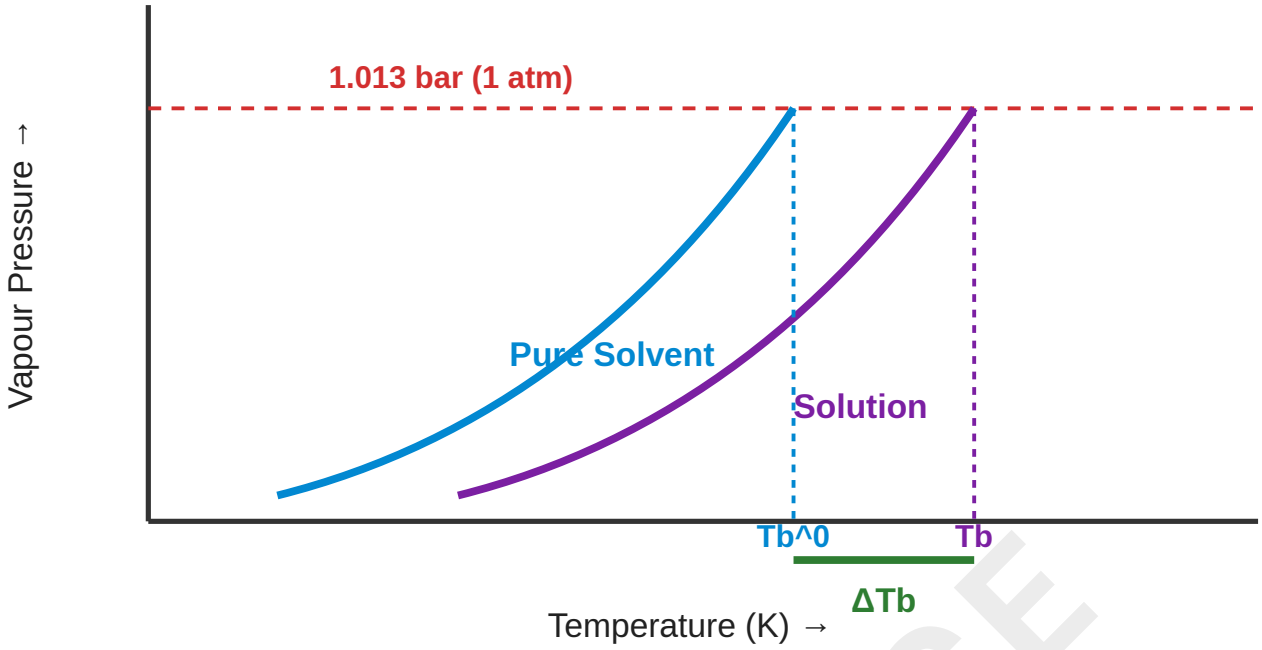
For dilute solutions where $n_2 \ll n_1$:

$$\frac{p_1^\circ - p_1}{p_1^\circ} \approx \frac{n_2}{n_1} = \frac{w_2 \times M_1}{M_2 \times w_1} \implies M_2 = \frac{w_2 \times M_1 \times p_1^\circ}{(p_1^\circ - p_1) \times w_1}$$

6.2 Elevation of Boiling Point (ΔT_b)

A liquid boils at the temperature at which its vapour pressure reaches external atmospheric pressure ($1.013\,\text{bar}$ or $1\,\text{atm}$). Because a non-volatile solute lowers the vapour pressure, the solution must be heated to a higher temperature to reach atmospheric pressure.

BOSONANCE



Tutorial Diagram 6: Vapour pressure curves for pure solvent and solution illustrating boiling point elevation ($\Delta T_\mathrm{b} = T_\mathrm{b} - T_\mathrm{b}^\circ$).

For dilute solutions, the elevation of boiling point is directly proportional to molality:

$$\Delta T_\mathrm{b} = T_\mathrm{b} - T_\mathrm{b}^\circ = K_\mathrm{b}\, m = \frac{K_\mathrm{b} \times 1000 \times w_2}{M_2 \times w_1}$$

$$M_2 = \frac{K_\mathrm{b} \times 1000 \times w_2}{\Delta T_\mathrm{b} \times w_1}$$

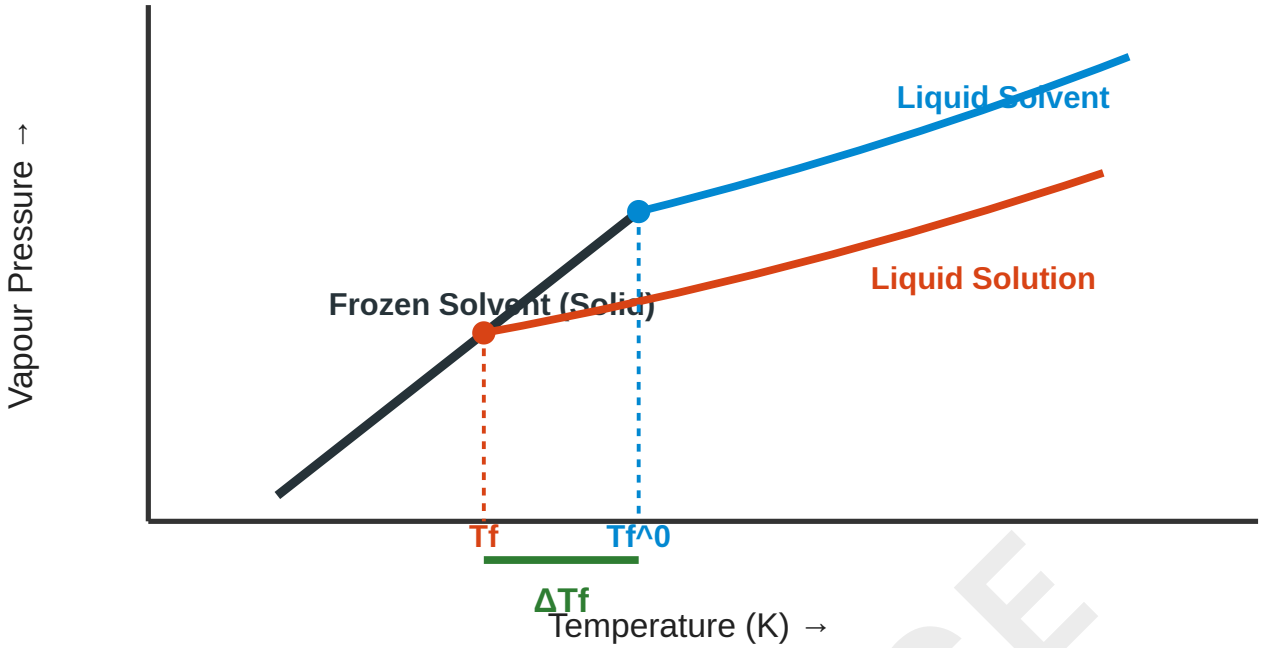
K_b is the **Ebullioscopic Constant** or **Molal Elevation Constant** (units: $\mathrm{K\,kg\,mol^{-1}}$). It is a solvent-specific intrinsic property calculated from:

$$K_\mathrm{b} = \frac{R\, M_1\, (T_\mathrm{b}^\circ)^2}{1000 \times \Delta_\mathrm{vap} H}$$

6.3 Depression of Freezing Point (ΔT_f)

The **freezing point** is the temperature at which the vapour pressure of the liquid phase equals the vapour pressure of its pure solid phase. Dissolving a non-volatile solute depresses the vapour pressure, forcing the solution curve to intersect the solid solvent curve at a lower temperature.

BOSONANCE



Tutorial Diagram 7: Freezing point depression diagram showing intersection with the frozen solid solvent curve ($\Delta T_\mathrm{f} = T_\mathrm{f}^\circ - T_\mathrm{f}$).

$$\Delta T_\mathrm{f} = T_\mathrm{f}^\circ - T_\mathrm{f} = K_\mathrm{f}\, m = \frac{K_\mathrm{f} \times 1000 \times w_2}{M_2 \times w_1}$$

$$M_2 = \frac{K_\mathrm{f} \times 1000 \times w_2}{\Delta T_\mathrm{f} \times w_1}$$

K_f is the **Cryoscopic Constant** or **Molal Depression Constant** (units: $\mathrm{K\,kg\,mol^{-1}}$):

$$K_\mathrm{f} = \frac{R\, M_1\, (T_\mathrm{f}^\circ)^2}{1000 \times \Delta_{\mathrm{fus}} H}$$

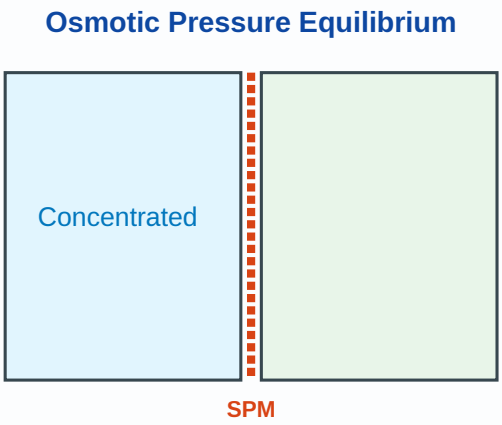
Tutorial Table 4: Boiling Point Elevation and Freezing Point Depression Constants

Solvent	Normal b.p. (K)	K_b ($\mathrm{K\,kg\,mol^{-1}}$)	Normal f.p. (K)	K_f ($\mathrm{K\,kg\,mol^{-1}}$)
Water ($\mathrm{H_2O}$)	373.15	0.52	273.15	1.86
Ethanol	351.5	1.20	155.7	1.99
Benzene	353.3	2.53	278.6	5.12
Cyclohexane	353.74	2.79	279.55	20.00
Carbon tetrachloride	350.0	5.03	250.5	31.8
Chloroform	334.4	3.63	209.6	4.79
Acetic acid	391.1	2.93	290.0	3.90

6.4 Osmosis and Osmotic Pressure (Π)

Osmosis is the spontaneous, net directional transport of solvent molecules from a region of pure solvent (or dilute solution) into a more concentrated solution across a **Semipermeable Membrane (SPM)**.
Osmotic Pressure (Π) is the exact hydrostatic pressure that must be applied to the solution compartment to halt the influx of solvent molecules across the SPM.

BOSONANCE



Solution Pure Solvent Applied P = Π + P_atm P_atm Net solvent transport halted: P = Π Reverse Osmosis (Desalination) Cellulose Acetate SPM Saline / Sea Water Fresh Potable Water Applied P > Π Pure Water Flow → Pure water forced through membrane

Tutorial Diagram 8: Experimental demonstration of Osmotic Pressure (Π) and Reverse Osmosis desalination using a selective cellulose acetate membrane.

For dilute solutions, van 't Hoff established that osmotic pressure obeys the equation:

$$\Pi = CRT = \left(\frac{n_2}{V}\right)RT = \frac{w_2 RT}{M_2 V} \implies M_2 = \frac{w_2 RT}{\Pi V}$$

Why Osmotic Pressure is Superior for Macromolecules:

- **Large, Measurable Magnitudes:** Even at micromolar concentrations ($10^{-4}\,\mathrm{M}$), Π generates measurable hydrostatic head heights, whereas ΔT_f or ΔT_b values are negligibly tiny ($\approx 10^{-4}\,\mathrm{K}$).
- **Ambient Temperature Measurement:** Measured around room temperature ($298\,\mathrm{K}$), preventing thermal denaturation of proteins and nucleic acids.
- **Molarity Based:** Uses molarity (C), which is straightforwardly prepared in volumetric glassware.

Isotonic, Hypertonic, and Hypotonic Solutions:

- **Isotonic Solutions:** Two solutions with identical osmotic pressures ($\Pi_1 = \Pi_2$) across an SPM at a given temperature; no net osmosis occurs.
- **Normal Saline:** Pure aqueous 0.9% (w/v) NaCl is isotonic with human red blood cell (RBC) intracellular fluid. Safe for intravenous administration.
- **Hypertonic Solution:** Salt concentration $> 0.9\%$ (w/v). Placed in hypertonic media, water exits the RBC via exosmosis, causing cell collapse and shrinkage (**crenation**).
- **Hypotonic Solution:** Salt concentration $< 0.9\%$ (w/v). Water floods into RBCs via endosmosis, causing swelling and eventual bursting (**haemolysis**).
- **Edema:** Excessive salt intake causes osmotic water retention in interstitial tissues, resulting in swollen, puffy tissues.
- **Food Preservation:** Salting meat and sugaring candied fruits exerts high osmotic pressure; bacteria landing on them rapidly lose cellular water through exosmosis, shrivel, and perish.

6.5 Reverse Osmosis and Desalination

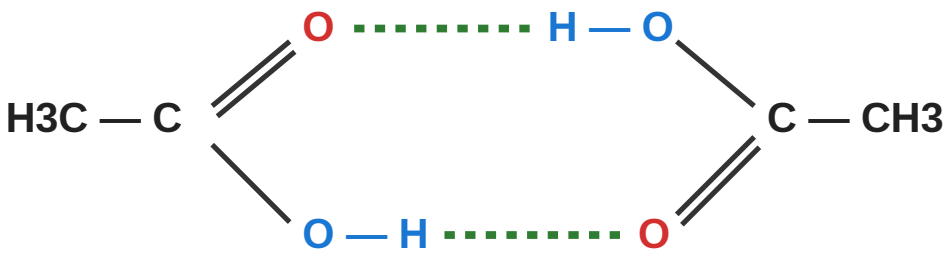
If external hydrostatic pressure exceeding the osmotic pressure ($P > \Pi$) is exerted upon the solution compartment, the natural osmotic flow is reversed: pure solvent molecules are squeezed backward through the SPM out of the saline solution into the freshwater compartment. Modern desalination installations employ synthetic, defect-free **cellulose acetate** membranes mounted on porous supports.

7. Abnormal Molar Masses and the van 't Hoff Factor (i)

When an electrolyte or polar molecule is dissolved in a solvent, it may undergo **dissociation** into ions or **association** (dimerization/polymerization) into multimers. This changes the actual number of dissolved particles:

- **Dissociation:** Number of solute particles increases $\implies$ Colligative property increases $\implies$ Experimentally determined molar mass is *lower* than normal.
- **Association:** Number of solute particles decreases $\implies$ Colligative property decreases $\implies$ Experimentally determined molar mass is *higher* than normal.

BOSONANCE



Intermolecular Hydrogen Bonds

Tutorial Diagram 9: Planar cyclic dimer of acetic acid (CH_3COOH) in non-polar solvents (e.g., benzene) stabilizing abnormal double molar mass.

7.1 Mathematical Formulations of the van 't Hoff Factor

$$i = \frac{\text{Normal (theoretical) molar mass}}{\text{Abnormal (observed) molar mass}} = \frac{\text{Observed colligative property}}{\text{Calculated colligative property (assuming no dissociation/association)}}$$

$$i = \frac{\text{Total moles of particles after dissociation/association}}{\text{Initial moles of particles before dissociation/association}}$$

Modified Colligative Equations:

- **RLVP:** $\dfrac{p_1^\circ - p_1}{p_1^\circ} = i\,x_2 \approx i\,\dfrac{n_2}{n_1}$
- **Boiling Point Elevation:** $\Delta T_\mathrm{b} = i\,K_\mathrm{b}\,m$
- **Freezing Point Depression:** $\Delta T_\mathrm{f} = i\,K_\mathrm{f}\,m$
- **Osmotic Pressure:** $\Pi = i\,C\,R\,T$

7.2 Relation Between i and Degree of Dissociation (α)

Consider 1 mole of an electrolyte A_xB_y which dissociates to produce n ions:

$$A_xB_y \rightleftharpoons xA^{y+} + yB^{x-} \quad (n = x + y)$$

At equilibrium:
• Undissociated moles = $1 - \alpha$
• Moles of ions produced = $n\alpha$
• Total moles at equilibrium = $(1 - \alpha) + n\alpha = 1 + (n - 1)\alpha$

$$i = \frac{1 + (n-1)\alpha}{1} \implies \alpha = \frac{i - 1}{n - 1}$$

7.3 Relation Between i and Degree of Association (α)

Consider n molecules associating into a multimer:

$$nA \rightleftharpoons A_n$$

At equilibrium:
• Unassociated moles = $1 - \alpha$
• Associated multimer moles = $\frac{\alpha}{n}$
• Total moles at equilibrium = $1 - \alpha + \frac{\alpha}{n}$

$$i = 1 - \alpha\left(1 - \frac{1}{n}\right) \implies \alpha = \frac{1 - i}{1 - \frac{1}{n}}$$

For dimerization (e.g., benzoic acid or acetic acid in benzene, $n = 2$):

$$i = 1 - \frac{\alpha}{2} \implies \alpha = 2(1 - i)$$

Tutorial Table 5: Experimental van 't Hoff Factors (i) of Strong Electrolytes at Differing Dilutions

Electrolyte	i at $0.1\,m$	i at $0.01\,m$	i at $0.001\,m$	Theoretical Complete Dissociation ($\alpha = 1$)
$NaCl$	1.87	1.94	1.97	2.00

BOSONANCE

Electrolyte	i at $0.1\,m$	i at $0.01\,m$	i at $0.001\,m$	Theoretical Complete Dissociation ($\alpha = 1$)
KCl	1.85	1.94	1.98	2.00
$MgSO_4$	1.21	1.53	1.82	2.00
K_2SO_4	2.32	2.70	2.84	3.00

8. Textbook Worked Examples with Complete Derivations

EXAMPLE 1: HENRY'S LAW MILLIMOLE CALCULATION

Problem: If N_2 gas is bubbled through water at $293\,K$, how many millimoles of N_2 gas will dissolve in 1 litre of water? Assume N_2 exerts a partial pressure of $0.987\,bar$. Given Henry's law constant for N_2 at $293\,K$ is $76.48\,kbar$.

Worked Reasoning:

1. Express K_H in matching units: $K_H = 76.48\,kbar = 76,480\,bar$.
2. Apply Henry's law:
$$x_{N_2} = \frac{p_{N_2}}{K_H} = \frac{0.987\,bar}{76,480\,bar} = 1.29 \times 10^{-5}$$
3. $1\,L$ of water weighs $1000\,g$. Moles of water:
$$n_{H_2O} = \frac{1000\,g}{18\,g\,mol^{-1}} = 55.5\,mol$$
4. Since $n_{N_2} \ll 55.5$:
$$x_{N_2} \approx \frac{n_{N_2}}{55.5} \implies n_{N_2} = 1.29 \times 10^{-5} \times 55.5 = 7.16 \times 10^{-4}\,mol$$
$$\text{Millimoles of } N_2 = 7.16 \times 10^{-4} \times 1000 = 0.716\,mmol$$

EXAMPLE 2: LIQUID-LIQUID VAPOUR & CONDENSATE COMPOSITION

Problem: Vapour pressures of chloroform ($CHCl_3$) and dichloromethane (CH_2Cl_2) at $298\,K$ are $200\,mm\,Hg$ and $415\,mm\,Hg$ respectively.
(i) Calculate the vapour pressure of the solution prepared by mixing $25.5\,g$ of $CHCl_3$ and $40.0\,g$ of CH_2Cl_2 at $298\,K$.
(ii) Calculate the mole fractions of each component in the equilibrium vapour phase.

Worked Reasoning:

1. Molar masses:
$CH_2Cl_2 = 12 + 2(1) + 2(35.5) = 85\,g\,mol^{-1}$.
$CHCl_3 = 12 + 1 + 3(35.5) = 119.5\,g\,mol^{-1}$.
2. Moles present in solution:
$$n_{CH_2Cl_2} = \frac{40.0}{85} = 0.470\,mol, \quad n_{CHCl_3} = \frac{25.5}{119.5} = 0.213\,mol$$
$$n_{total} = 0.470 + 0.213 = 0.683\,mol$$
3. Mole fractions in liquid solution:
$$x_{CH_2Cl_2} = \frac{0.470}{0.683} = 0.688, \quad x_{CHCl_3} = 1 - 0.688 = 0.312$$
4. Partial vapour pressures:
$$p_{CH_2Cl_2} = x_{CH_2Cl_2}\,p^\circ_{CH_2Cl_2} = 0.688 \times 415\,mm\,Hg = 285.52\,mm\,Hg$$
$$p_{CHCl_3} = x_{CHCl_3}\,p^\circ_{CHCl_3} = 0.312 \times 200\,mm\,Hg = 62.40\,mm\,Hg$$
$$p_{total} = 285.52 + 62.40 = 347.92\,mm\,Hg$$
5. Vapour phase mole fractions:
$$y_{CH_2Cl_2} = \frac{285.52}{347.92} = 0.82 \quad \text{and} \quad y_{CHCl_3} = \frac{62.40}{347.92} = 0.18$$

Notice that the vapour phase is substantially enriched in the more volatile component (CH_2Cl_2, whose pure vapour pressure is $415\,mm\,Hg$).

BOSONANCE

EXAMPLE 3: BENZOIC ACID DIMERIZATION & PERCENTAGE ASSOCIATION

Problem: $2.0\,\text{g}$ of benzoic acid ($\text{C}_6\text{H}_5\text{COOH}$) dissolved in $25.0\,\text{g}$ of benzene exhibits a freezing point depression of $1.62\,\text{K}$. Given K_f for benzene is $4.9\,\text{K kg mol}^{-1}$, determine the percentage association of the acid into dimers in benzene.

Worked Reasoning:

1. Normal molar mass of benzoic acid ($\text{C}_7\text{H}_6\text{O}_2$):
$$M_\text{normal} = 7(12) + 6(1) + 2(16) = 122\,\text{g mol}^{-1}$$
2. Observed (experimental) molar mass from colligative data:
$$M_\text{obs} = \frac{K_\text{f} \times w_2 \times 1000}{\Delta T_\text{f} \times w_1} = \frac{4.9 \times 2.0 \times 1000}{1.62 \times 25.0} = 241.98\,\text{g mol}^{-1}$$
3. Determine van 't Hoff factor i:
$$i = \frac{M_\text{normal}}{M_\text{obs}} = \frac{122}{241.98} = 0.504$$
4. Relate i to degree of association (α) for a dimer ($n = 2$):
$$2\,\text{C}_6\text{H}_5\text{COOH} \rightleftharpoons (\text{C}_6\text{H}_5\text{COOH})_2$$
$$i = 1 - \frac{\alpha}{2} \implies \frac{\alpha}{2} = 1 - i = 1 - 0.504 = 0.496 \implies \alpha = 0.992$$

Thus, benzoic acid is 99.2% associated into dimers.

9. High-Yield NEET UG Practice Questions with Worked Solutions

Practice Question 1 • Single Option Correct

At a specified temperature, the Henry's law constants (K_H) in water for gases P, Q, and R are $144.97\,\text{kbar}$, $69.16\,\text{kbar}$, and $0.413\,\text{kbar}$ respectively. Which of the following arrangements correctly denotes the order of increasing solubility of these gases under identical partial pressures?

(A) $P < Q < R$

(B) $R < Q < P$

(C) $Q < P < R$

(D) $R < P < Q$

Correct Answer: (A)

Worked Solution: According to Henry's Law, $p = K_\text{H} x$, or $x = p/K_\text{H}$. At constant partial pressure p, solubility (measured by mole fraction x) is inversely proportional to K_H. The given values are:
$K_\text{H}(P) = 144.97\,\text{kbar}$ (highest $K_\text{H} \implies$ lowest solubility)
$K_\text{H}(Q) = 69.16\,\text{kbar}$
$K_\text{H}(R) = 0.413\,\text{kbar}$ (lowest $K_\text{H} \implies$ highest solubility)
Therefore, the order of increasing solubility is $P < Q < R$.

BOSONANCE

Practice Question 2 • Single Option Correct

Two volatile liquids A and B form an ideal solution. At $300\ \mathrm{K}$, the vapour pressures of pure A and pure B are $200\ \mathrm{mm\ Hg}$ and $100\ \mathrm{mm\ Hg}$ respectively. If an equimolar liquid mixture of A and B is allowed to equilibrate with its vapour, the mole fraction of component A in the vapour phase (y_A) is:

(A) 0.33

(B) 0.50

(C) 0.67

(D) 0.75

Correct Answer: (C)

Worked Solution: For an equimolar liquid mixture, $x_A = 0.5$ and $x_B = 0.5$.
Partial pressures:
$p_A = x_A p_A^\circ = 0.5 \times 200 = 100\ \mathrm{mm\ Hg}$
$p_B = x_B p_B^\circ = 0.5 \times 100 = 50\ \mathrm{mm\ Hg}$
Total vapour pressure: $p_{\text{total}} = 100 + 50 = 150\ \mathrm{mm\ Hg}$
Mole fraction of A in vapour:

$$y_A = \frac{p_A}{p_{\text{total}}} = \frac{100}{150} = \frac{2}{3} \approx 0.67$$

Practice Question 3 • Statement Evaluation

Consider the following statements regarding a mixture of chloroform and acetone:
Statement I: The mixture shows negative deviation from Raoult's Law because of hydrogen bond formation between the chloroform proton and the carbonyl oxygen of acetone.
Statement II: The mixture undergoes endothermic mixing with an overall increase in solution volume ($\Delta_{\text{mix}}H > 0$ and $\Delta_{\text{mix}}V > 0$).
Select the correct option:

(A) Both Statement I and Statement II are correct.

(B) Both Statement I and Statement II are incorrect.

(C) Statement I is correct, but Statement II is incorrect.

(D) Statement I is incorrect, but Statement II is correct.

Correct Answer: (C)

Worked Solution:
• Statement I is correct: Acetone and chloroform form a strong intermolecular hydrogen bond ($\mathrm{Cl_3C\text{--}H\cdots O=C(CH_3)_2}$) which is stronger than the solute-solute or solvent-solvent dipole interactions, resulting in negative deviation from Raoult's Law.
• Statement II is incorrect: For negative deviations, new attractive forces are formed, releasing energy ($\Delta_{\text{mix}}H < 0$, exothermic) and pulling molecules closer together ($\Delta_{\text{mix}}V < 0$, contraction).

BOSONANCE

Practice Question 4 • Assertion & Reason

Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R):
Assertion (A): When human red blood cells are suspended in a 0.2% (w/v) aqueous NaCl solution, they swell and eventually undergo haemolysis.
Reason (R): A 0.2% (w/v) NaCl solution is hypotonic with respect to the intracellular fluid of red blood cells, causing a net osmotic influx of water into the cells.

(A) Both (A) and (R) are true and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is NOT the correct explanation of (A).

(C) (A) is true, but (R) is false.

(D) (A) is false, but (R) is true.

Correct Answer: (A)

Worked Solution: The intracellular fluid of RBCs is isotonic with 0.9% (w/v) NaCl. A 0.2% (w/v) NaCl solution has lower osmotic pressure (hypotonic). Consequently, water flows through the semipermeable cell membrane from the exterior into the RBCs via endosmosis, causing swelling and eventual bursting (haemolysis). Both (A) and (R) are true, and (R) correctly explains (A).

Practice Question 5 • Single Option Correct

Which of the following aqueous solutions will exhibit the lowest freezing point? (Assume complete dissociation of all ionic salts)

(A) $0.10\,m$ Glucose

(B) $0.05\,m\,\mathrm{K_2SO_4}$

(C) $0.06\,m\,\mathrm{NaCl}$

(D) $0.08\,m\,\mathrm{Al_2(SO_4)_3}$

Correct Answer: (D)

Worked Solution: The depression in freezing point is $\Delta T_\mathrm{f} = i\,K_\mathrm{f}\,m$. The solution with the largest value of $(i \times m)$ experiences the greatest depression, resulting in the *lowest freezing point*:
• Glucose: non-electrolyte, $i = 1 \implies i \times m = 1 \times 0.10 = 0.10\,\mathrm{mol\,kg^{-1}}$
• $\mathrm{K_2SO_4}$: dissociates into $2\mathrm{K^+} + \mathrm{SO_4^{2-}}$ $(n = 3, i = 3) \implies i \times m = 3 \times 0.05 = 0.15\,\mathrm{mol\,kg^{-1}}$
• NaCl: dissociates into $\mathrm{Na^+} + \mathrm{Cl^-}$ $(n = 2, i = 2) \implies i \times m = 2 \times 0.06 = 0.12\,\mathrm{mol\,kg^{-1}}$
• $\mathrm{Al_2(SO_4)_3}$: dissociates into $2\mathrm{Al^{3+}} + 3\mathrm{SO_4^{2-}}$ $(n = 5, i = 5) \implies i \times m = 5 \times 0.08 = 0.40\,\mathrm{mol\,kg^{-1}}$
Since $\mathrm{Al_2(SO_4)_3}$ produces the largest effective particle concentration $(0.40\,\mathrm{mol\,kg^{-1}})$, it lowers the freezing point the most and has the lowest freezing point.

BOSONANCE

Practice Question 6 • Matching Type

Match the solution pairs listed in Column I with the appropriate thermodynamic / deviation characteristics in Column II:

Column I (Binary System)	Column II (Solution Property)
(P) n-hexane + n-heptane	(1) Forms maximum boiling azeotrope; $\Delta_{\text{mix}}H < 0$
(Q) Ethanol + Acetone	(2) Ideal solution; $\Delta_{\text{mix}}H = 0, \Delta_{\text{mix}}V = 0$
(R) Nitric acid (68%) + Water (32%)	(3) Minimum boiling azeotrope; $\Delta_{\text{mix}}H > 0$
(S) Ethanoic acid in benzene	(4) Solute dimerization; van 't Hoff factor $i < 1$

(A) P-2, Q-3, R-1, S-4

(B) P-1, Q-2, R-3, S-4

(C) P-2, Q-1, R-3, S-4

(D) P-3, Q-2, R-1, S-4

Correct Answer: (A)

Worked Solution:

• (P) n-hexane + n-heptane have nearly identical structural forces, forming an *ideal solution* ($\Delta_{\text{mix}}H = 0, \Delta_{\text{mix}}V = 0$) $\implies$ (2).

• (Q) Ethanol + Acetone shows *positive deviation* due to broken H-bonds, forming a minimum boiling azeotrope with $\Delta_{\text{mix}}H > 0$ $\implies$ (3).

• (R) $68\%\ \text{HNO}_3 + 32\%\ \text{H}_2\text{O}$ shows *negative deviation*, boiling at $393.5\,\text{K}$ (maximum boiling azeotrope) $\implies$ (1).

• (S) Ethanoic acid dimerizes in benzene via hydrogen bonding, cutting particle numbers in half ($i \approx 0.5 < 1$) $\implies$ (4).

Practice Question 7 • Single Option Correct

A $200\,\text{mL}$ aqueous solution contains $1.26\,\text{g}$ of a purified biomacromolecule. The osmotic pressure of this solution at $300\,\text{K}$ is found to be $2.57 \times 10^{-3}\,\text{bar}$. Using $R = 0.083\,\text{L bar K}^{-1}\,\text{mol}^{-1}$, the molar mass of the biomacromolecule is:

(A) $31,022\,\text{g mol}^{-1}$

(B) $61,022\,\text{g mol}^{-1}$

(C) $122,044\,\text{g mol}^{-1}$

(D) $15,511\,\text{g mol}^{-1}$

Correct Answer: (B)

Worked Solution: Using the osmotic pressure equation:

$$\Pi = \frac{w_2 RT}{M_2 V} \implies M_2 = \frac{w_2 RT}{\Pi V}$$

Given:

$w_2 = 1.26\,\text{g}$

$R = 0.083\,\text{L bar K}^{-1}\,\text{mol}^{-1}$

$T = 300\,\text{K}$

$V = 200\,\text{mL} = 0.200\,\text{L}$

$\Pi = 2.57 \times 10^{-3}\,\text{bar}$

$$M_2 = \frac{1.26 \times 0.083 \times 300}{(2.57 \times 10^{-3}) \times 0.200} = \frac{31.374}{5.14 \times 10^{-4}} = 61,038.9\,\text{g mol}^{-1} \approx 61,022\,\text{g mol}^{-1}$$

BOSONANCE

Practice Question 8 • Single Option Correct

A $0.0106\ \text{mol kg}^{-1}$ aqueous solution of a monobasic organic acid (HA) freezes at -0.0205°C. If the cryoscopic constant of water is $K_\text{f} = 1.86\ \text{K kg mol}^{-1}$, what is the degree of dissociation (α) of the acid?

(A) 0.010	(B) 0.041
(C) 0.082	(D) 0.104

Correct Answer: (B)

Worked Solution:

1. Observed freezing point depression: $\Delta T_{\text{f(obs)}} = 0.0205\ \text{K}$.
2. Theoretical freezing point depression (assuming no ionization):

$$\Delta T_{\text{f(calc)}} = K_\text{f} \times m = 1.86\ \text{K kg mol}^{-1} \times 0.0106\ \text{mol kg}^{-1} = 0.01971\ \text{K}$$

3. Determine van 't Hoff factor:

$$i = \frac{\Delta T_{\text{f(obs)}}}{\Delta T_{\text{f(calc)}}} = \frac{0.0205}{0.01971} = 1.040 \approx 1.041$$

4. For a monobasic acid ($\text{HA} \rightleftharpoons \text{H}^+ + \text{A}^-$, $n = 2$):

$$i = 1 + \alpha \implies \alpha = i - 1 = 1.041 - 1.000 = 0.041$$

Thus, the degree of dissociation is 0.041 (or 4.1%).

10. Common Pitfalls and High-Yield Revision Checklist

Avoid These Common NEET Mistakes

- **Confusing Molarity with Molality:** Remember that molarity uses total solution volume (L), while molality uses mass of pure *solvent* (kg), **not** total solution mass!
- **Neglecting i in Colligative Calculations:** Always check whether the solute is an electrolyte (e.g., NaCl, CaCl_2, $\text{K}_4[\text{Fe(CN)}_6]$) or associates (e.g., carboxylic acids in benzene). Omitting the van 't Hoff factor is the most frequent calculation mistake in NEET.
- **Freezing Point vs. Freezing Point Depression:** ΔT_f is the *lowering*: $\Delta T_\text{f} = T_\text{f}^\circ - T_\text{f}$. The actual freezing point of the solution is $T_\text{f} = T_\text{f}^\circ - \Delta T_\text{f}$. (For water, $0^\circ\text{C} - \Delta T_\text{f}$).
- **Azeotropes Cannot be Fractionally Distilled:** At the azeotropic composition, the liquid and vapour have identical compositions ($x_i = y_i$). They boil at a constant temperature like a single pure compound.
- **Units in Gas Constant (R):** When calculating osmotic pressure in bar, use $R = 0.083\ \text{L bar K}^{-1}\ \text{mol}^{-1}$. If osmotic pressure is in atm, use $R = 0.0821\ \text{L atm K}^{-1}\ \text{mol}^{-1}$. If Π is in Pa and volume in m^3, use $R = 8.314\ \text{J K}^{-1}\ \text{mol}^{-1}$.

Rapid NEET Revision Checklist

1. Solid solutions: H_2 in Pd (gas in solid); amalgam of Hg with Na (liquid in solid); dental fillings, alloys (solid in solid).
2. Temperature-independent concentration terms: Mass \%, Mole fraction (x), Molality (m), and ppm.
3. Henry's Law: $p = K_\text{H} x$; high $K_\text{H} \implies$ low solubility; K_H increases with temperature; gas dissolution in liquid is exothermic ($\Delta H < 0$).
4. Ideal Solution: $\Delta_\text{mix} H = 0$, $\Delta_\text{mix} V = 0$, $\Delta_\text{mix} S > 0$, $\Delta_\text{mix} G < 0$; A--$B \approx A$--$A \approx B$--B.
5. Positive Deviation: A--B weaker; $\Delta_\text{mix} H > 0$, $\Delta_\text{mix} V > 0$; forms minimum boiling azeotrope (e.g., ethanol + water, acetone + CS_2).
6. Negative Deviation: A--B stronger; $\Delta_\text{mix} H < 0$, $\Delta_\text{mix} V < 0$; forms maximum boiling azeotrope (e.g., chloroform + acetone, $\text{HNO}_3 + \text{H}_2\text{O}$).
7. Osmotic Pressure is preferred for biomolecules: runs at room temperature, directly yields molarity, gives large measurable values even at low solute concentrations.
8. van 't Hoff factor relations: $\alpha = \frac{i-1}{n-1}$ (dissociation); $\alpha = \frac{1-i}{1-1/n}$ (association).

BOSONANCE

Final Academic Summary

The physical behaviour of solutions is determined by the balance of intermolecular interactions between solute and solvent molecules. For ideal mixtures, interactions are uniform, following Raoult's Law across all proportions. Real mixtures deviate based on molecular affinities: breaking solvent associations causes positive deviation (higher vapour pressure, lower boiling points), whereas forming new associations (such as hydrogen bonds) causes negative deviation (lower vapour pressure, higher boiling points).

Colligative properties depend on the concentration of solute particles, not their identity. They provide reliable experimental methods to determine molecular masses, examine electrolyte dissociation, and measure degree of association. Keep these principles and formulas in mind as you prepare for the NEET UG examination.

BOSONANCE